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COURSE CODE: SBT-DF204-COMPUTER CASE STUDIES

LAB NAME: Case Study4 INVESTIGATING RHION POSSESSION EVIDENCE

DATE: 26/10/26

INSTRUCTOR: AMINU IDRIS

Computer Forensics Case Studies

Case Study 4 Investigating Rhion Possession Evidence

Student Assessment Brief

Assessment detail	Information
Course	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 4 Individual recovered-media and network-traffic investigation
Case title	Investigating Rhion Possession Evidence
Submission format	Written report with labelled screenshots, timeline, and evidence log
Deadline and maximum score	As published in E-Campus

This individual case study brings together disk-image recovery, steganography, FTP, HTTP, and Wireshark analysis. You will work from the authorised course evidence package and produce an evidence-led account of what the artefacts do and do not establish. Your report must clearly separate direct observations from inferences about file transfer, concealment, possession, or the person who used a device.

Case scenario

This is a fictional training investigation concerning imagery that may meet an illegal-possession threshold in the case scenario. The evidence package includes a seized USB disk image and network traces associated with FTP and HTTP activity. Investigators need to know whether deleted files, hidden data, reconstructed network objects, and the timing of recorded transfers form a defensible sequence. The task is an offline examination of supplied training artefacts; it does not authorise any contact with people, accounts, systems, hosts, or services named in the material.

Case materials

Use only the materials supplied through E-Campus and the accompanying teaching decks. The course sequence includes a USB DD image, an FTP trace (rhino.log), HTTP traces (rhino2.log and rhino3.log), and supporting recovery and Wireshark guidance. Use the exact filename in your evidence package if it differs

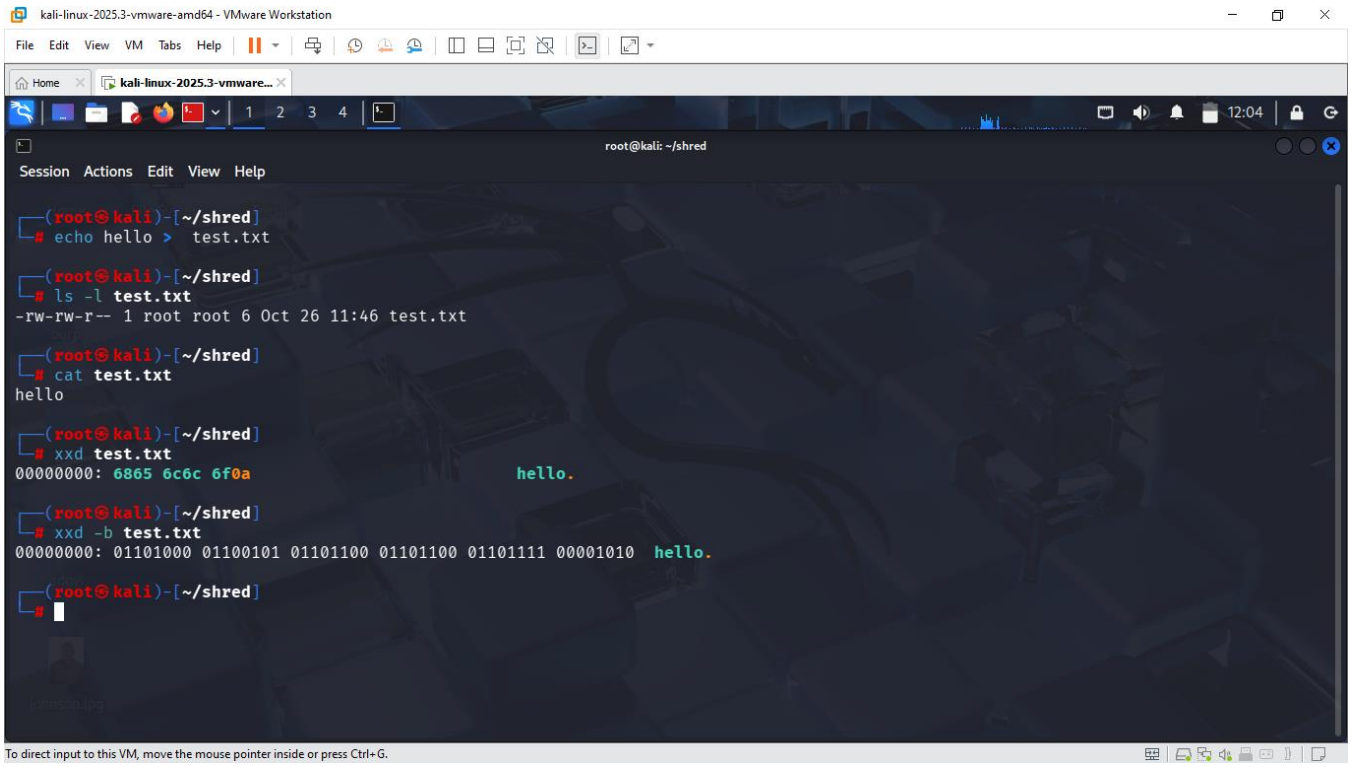
from the deck. Before analysis, create a working copy and record integrity information for every original file. Do not alter, shred, or dispose of original course material before your work has been assessed.

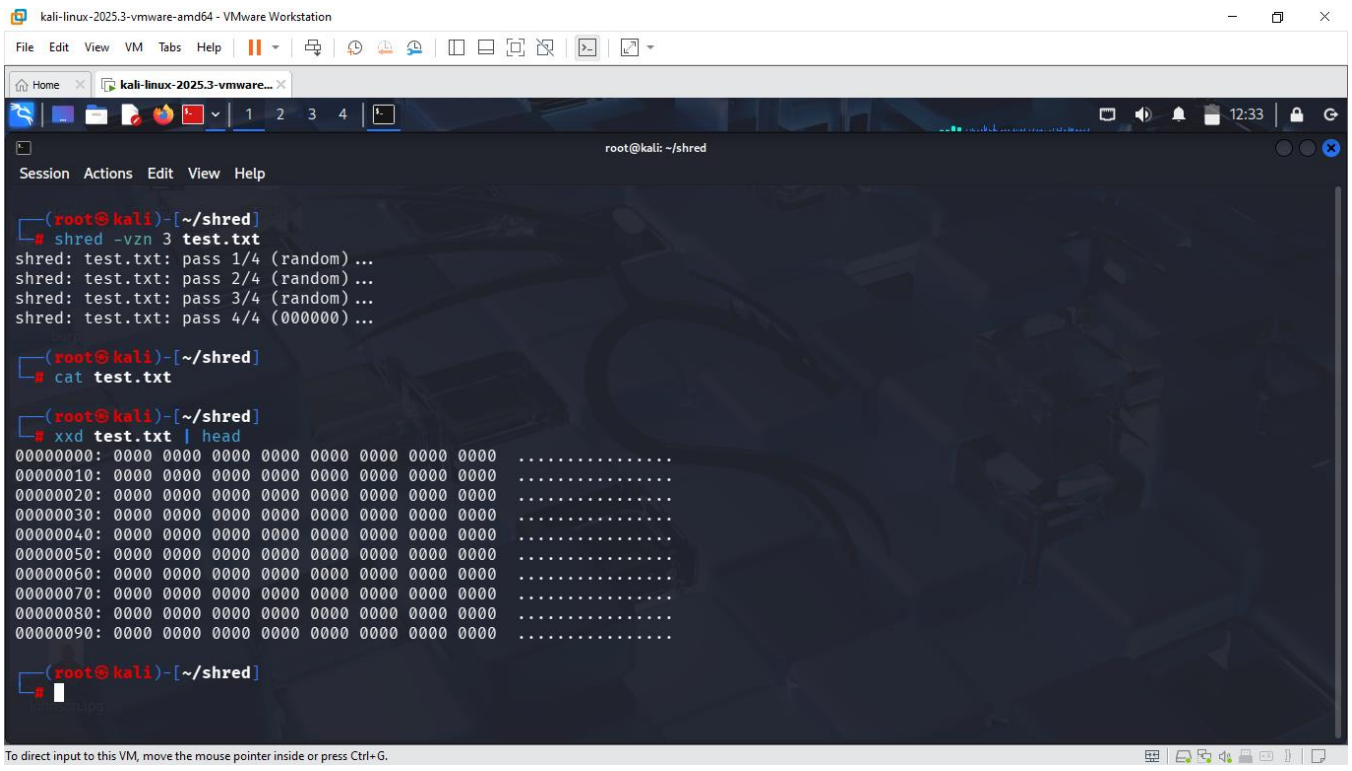
What you must submit

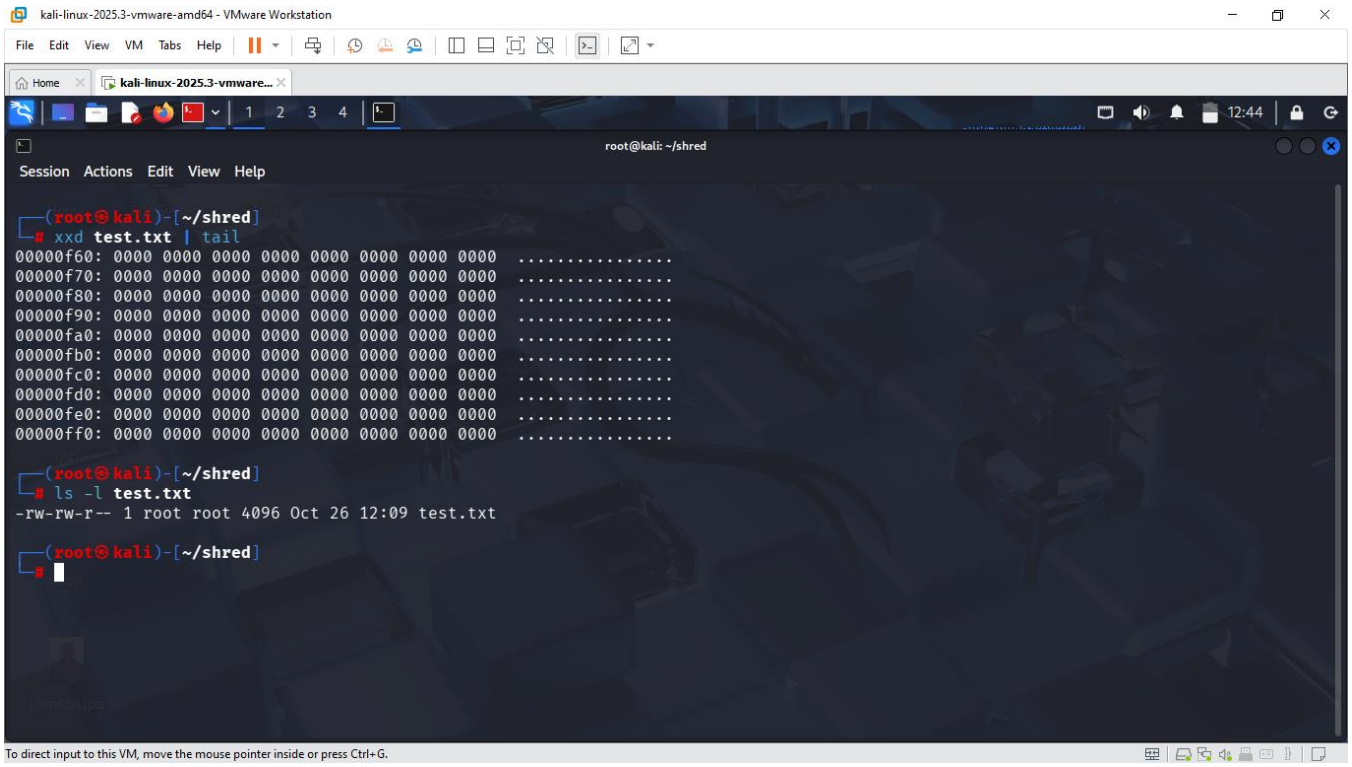
Item	Required output	What to provide
1	Forensic working set	Record the supplied disk image, packet captures, and any recovered artefact by original file name, byte size, SHA-256 value, source location, and working-copy location. Preserve the original course material unchanged.
2	Recovery and analysis record	Keep labelled output or screenshots for the disk-image layout, file recovery, metadata review, steganography testing, FTP and HTTP analysis, reconstructed objects, and validation checks.
3	Correlated timeline	Create a UTC, time-ordered account with at least five material events across at least three artefact families. Cite packet numbers or streams, file paths, hashes, timestamps, and evidence-log IDs.
4	Student report	Submit one report in PDF or DOCX format with a completed evidence log and timeline appendix. Include only the supporting files requested by your lecturer.

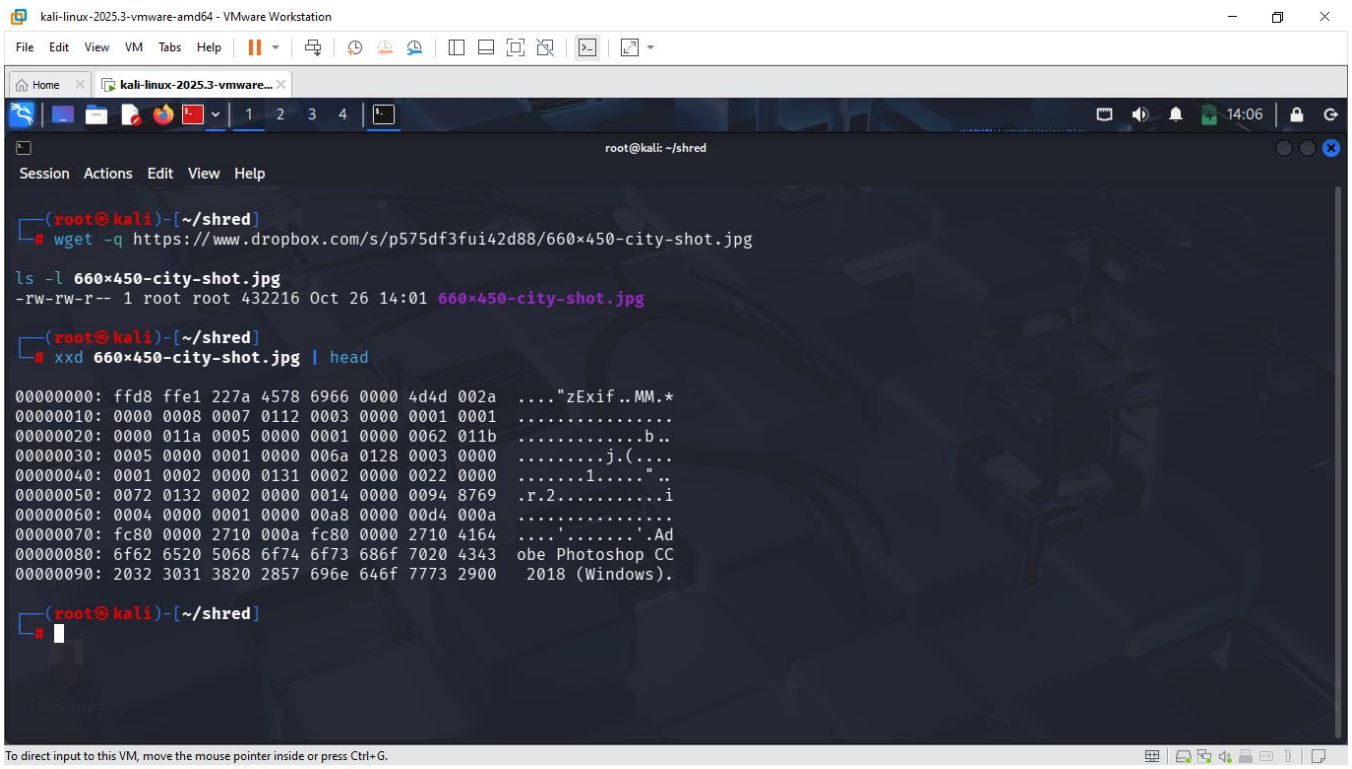
ANSWER



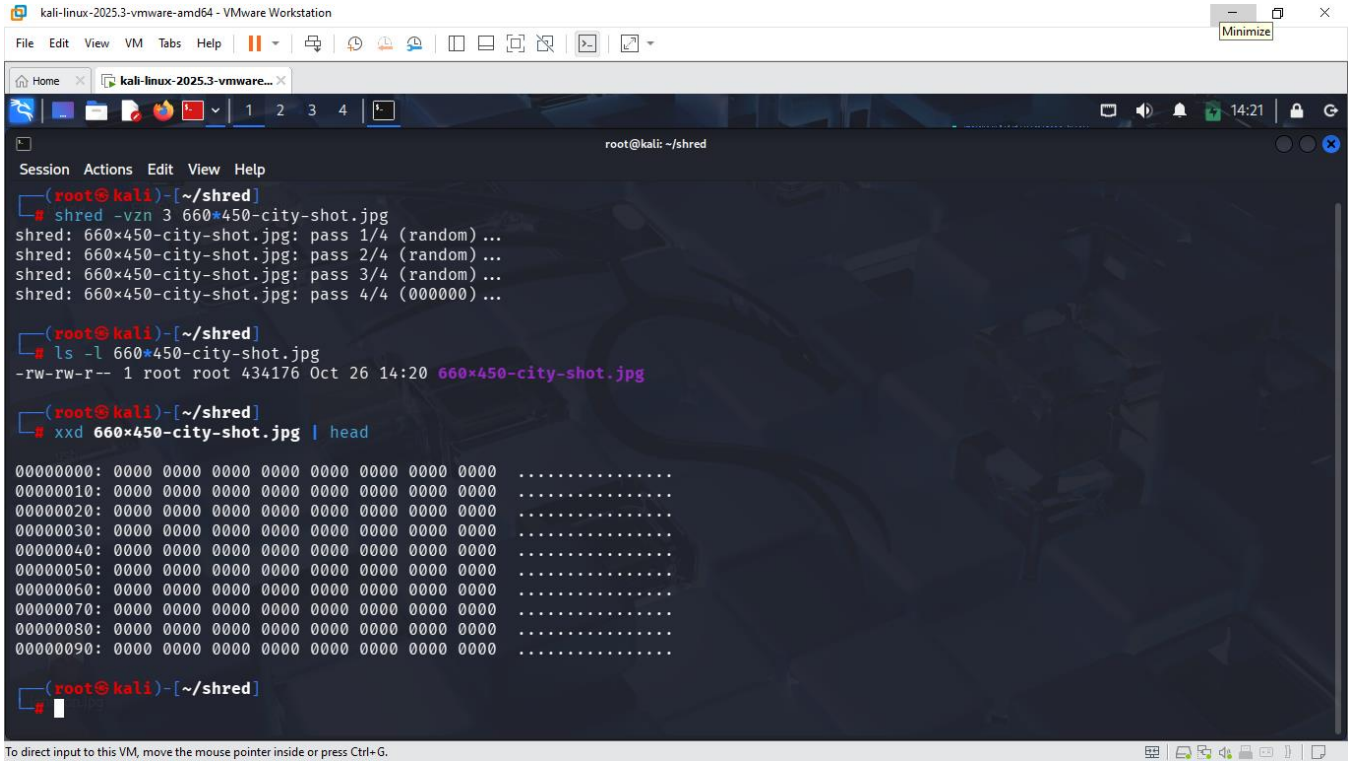

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.3-vmware...
12:04
root@kali: ~/shred
Session Actions Edit View Help
┌──(root㉿kali)-[~/shred]
└─# echo hello > test.txt
┌──(root㉿kali)-[~/shred]
└─# ls -l test.txt
-rw-rw-r-- 1 root root 6 Oct 26 11:46 test.txt
┌──(root㉿kali)-[~/shred]
└─# cat test.txt
hello
┌──(root㉿kali)-[~/shred]
└─# xxd test.txt
00000000: 6865 6c6c 6f0a hello.
┌──(root㉿kali)-[~/shred]
└─# xxd -b test.txt
00000000: 01101000 01100101 01101100 01101100 01101111 00001010 hello.
┌──(root㉿kali)-[~/shred]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.3-vmware...
12:33
root@kali: ~/shred
Session Actions Edit View Help
┌──(root㉿kali)-[~/shred]
└─# shred -vzn 3 test.txt
shred: test.txt: pass 1/4 (random)...
shred: test.txt: pass 2/4 (random)...
shred: test.txt: pass 3/4 (random)...
shred: test.txt: pass 4/4 (000000)...
┌──(root㉿kali)-[~/shred]
└─# cat test.txt
┌──(root㉿kali)-[~/shred]
└─# xxd test.txt | head
00000000: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000010: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000020: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000030: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000040: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000050: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000060: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000070: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000 ....
┌──(root㉿kali)-[~/shred]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



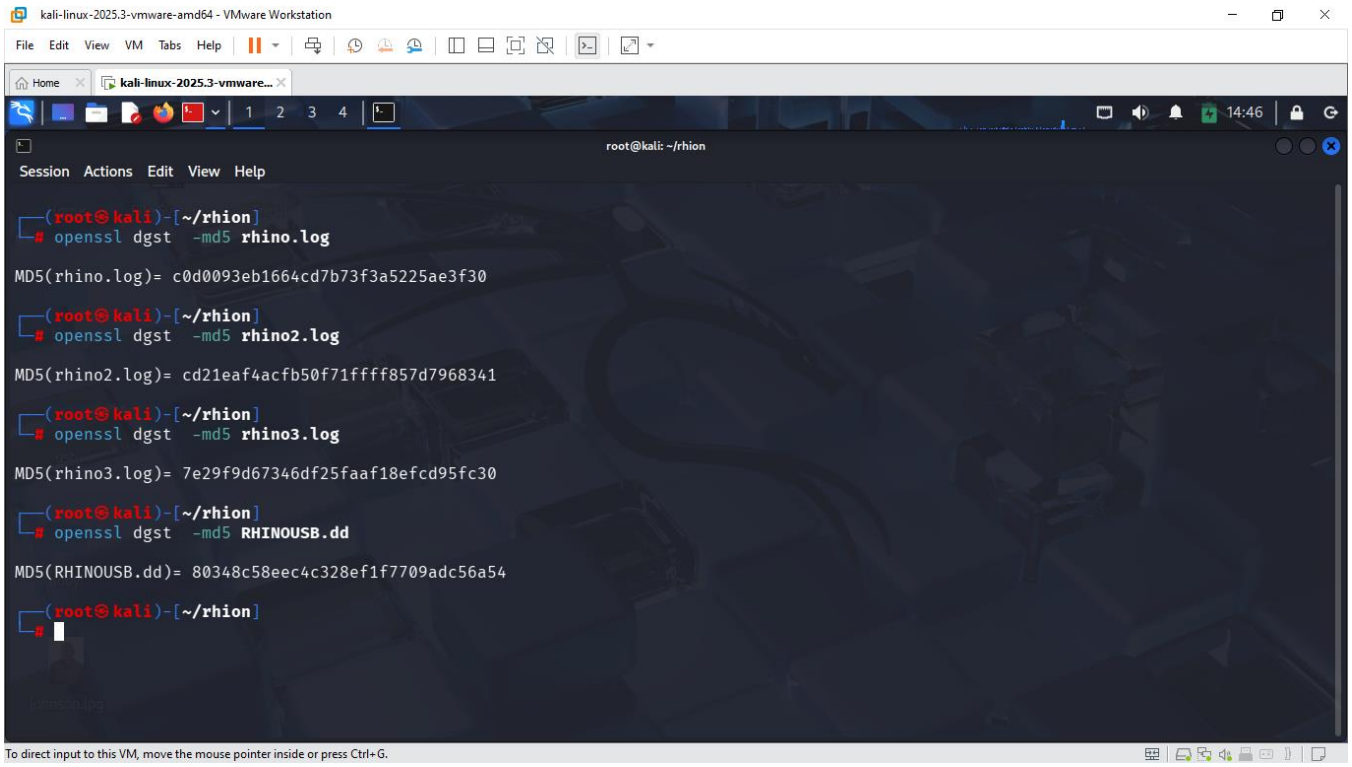


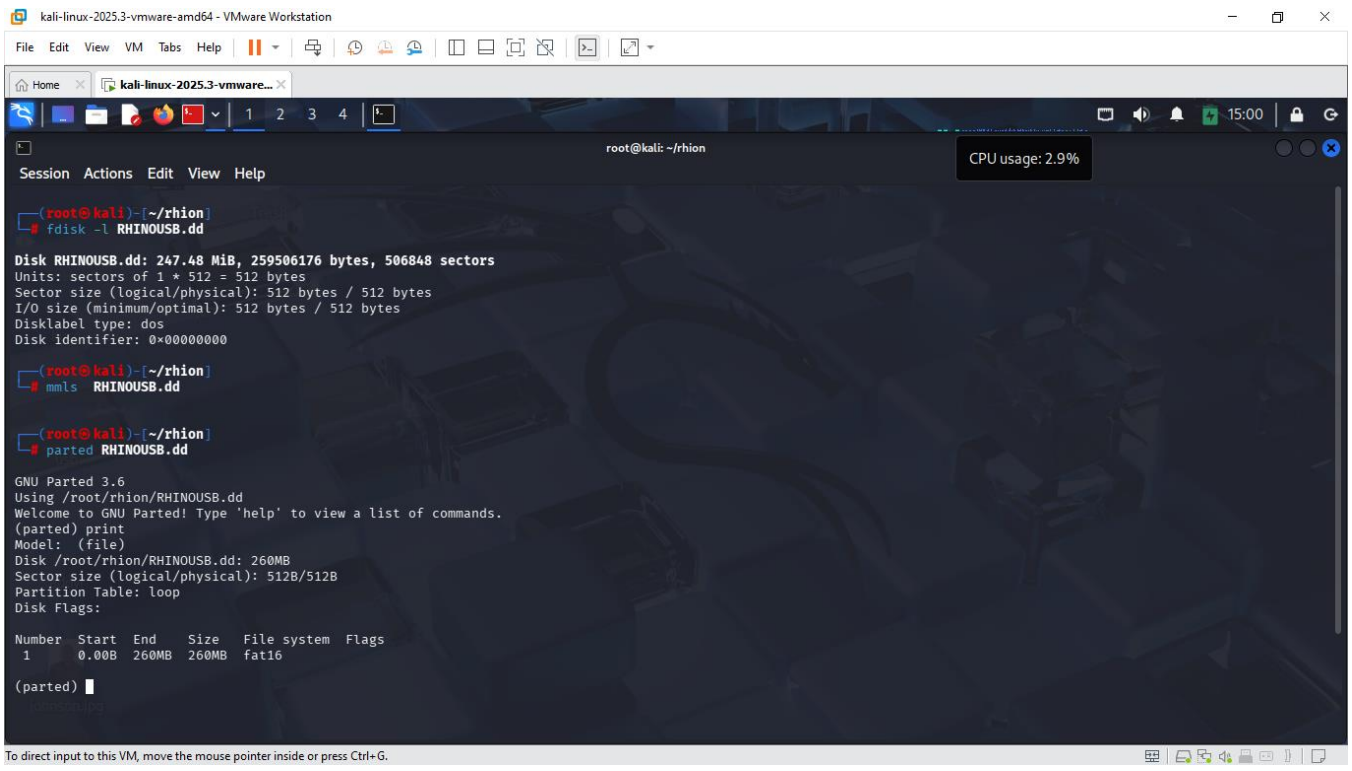
COMMAND USED: xxd 660x450-city-shot.jpg | head

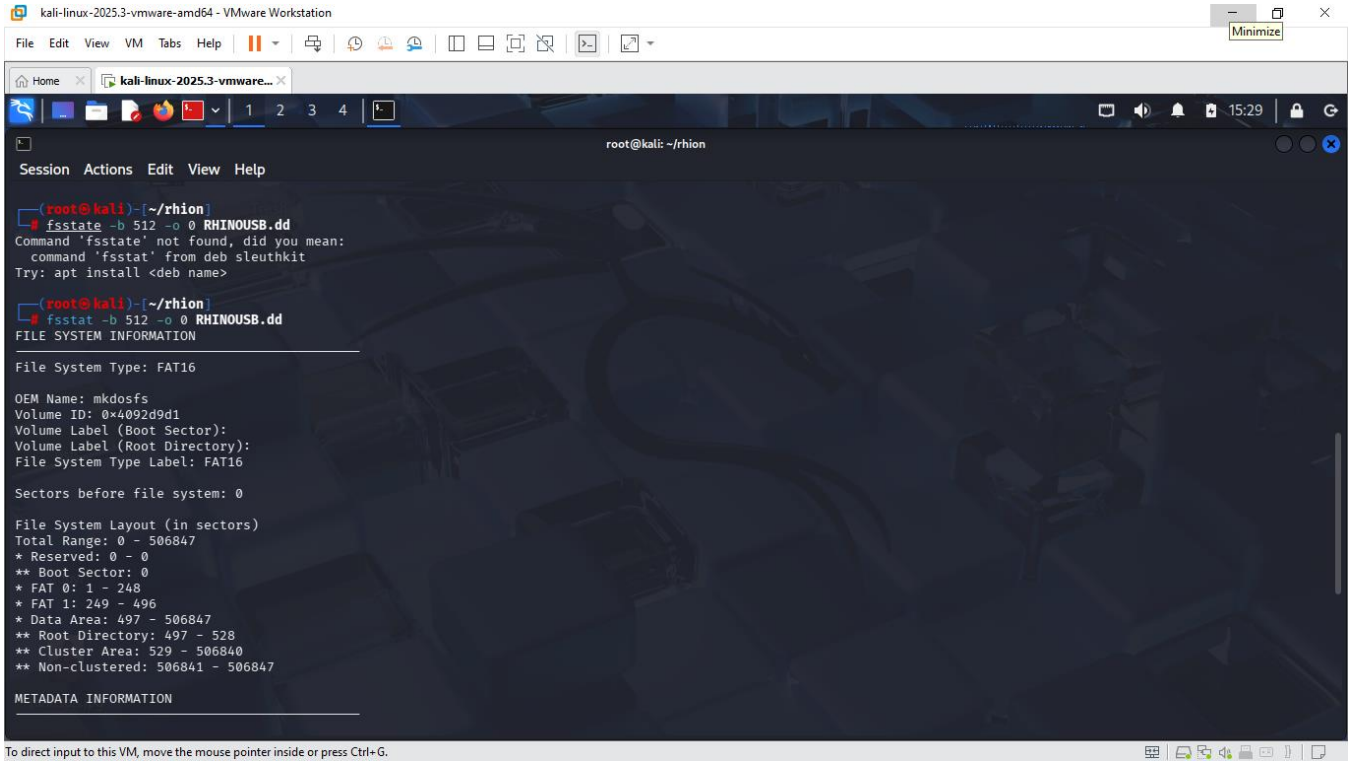


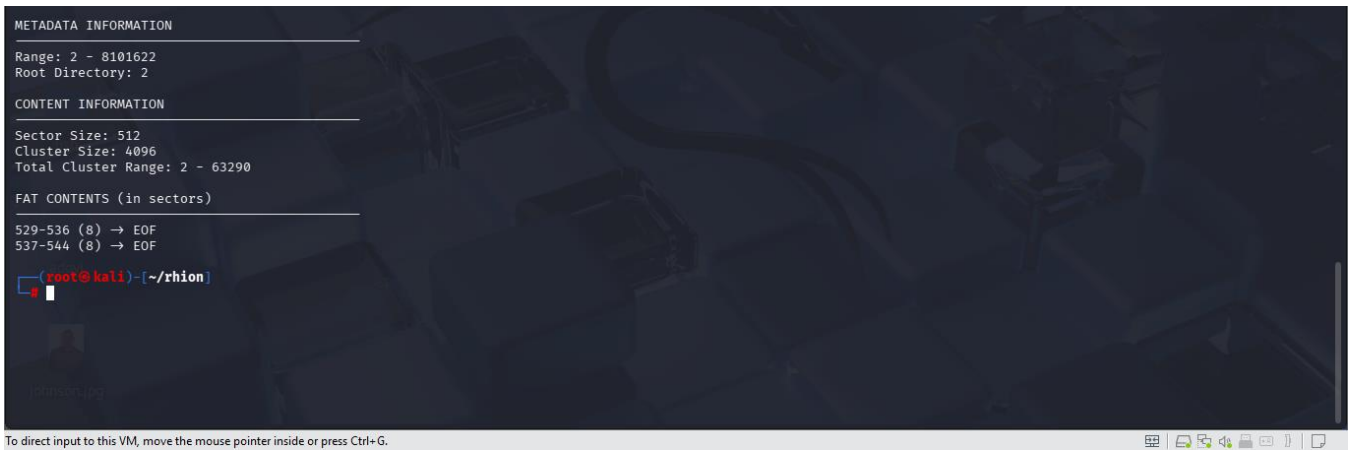


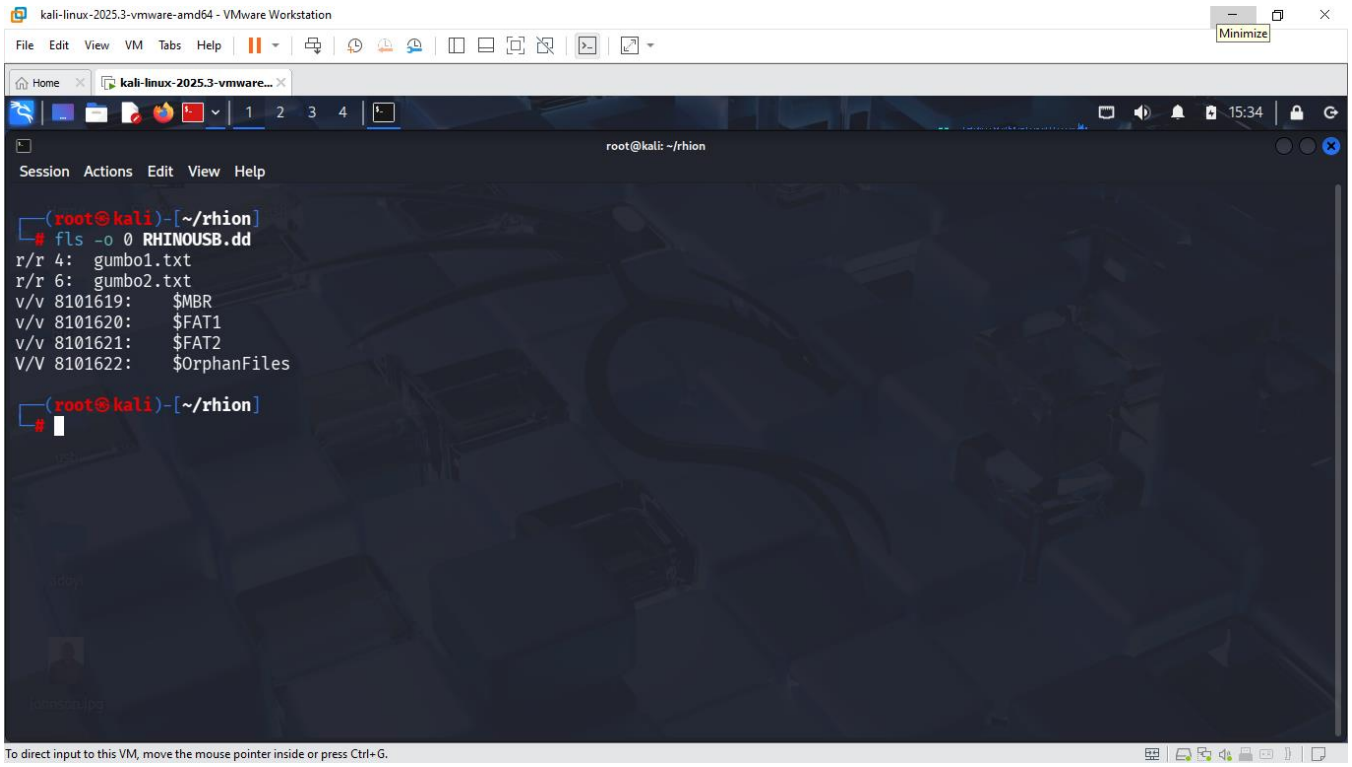
ANSWER: MD5 HASHES BELOW:

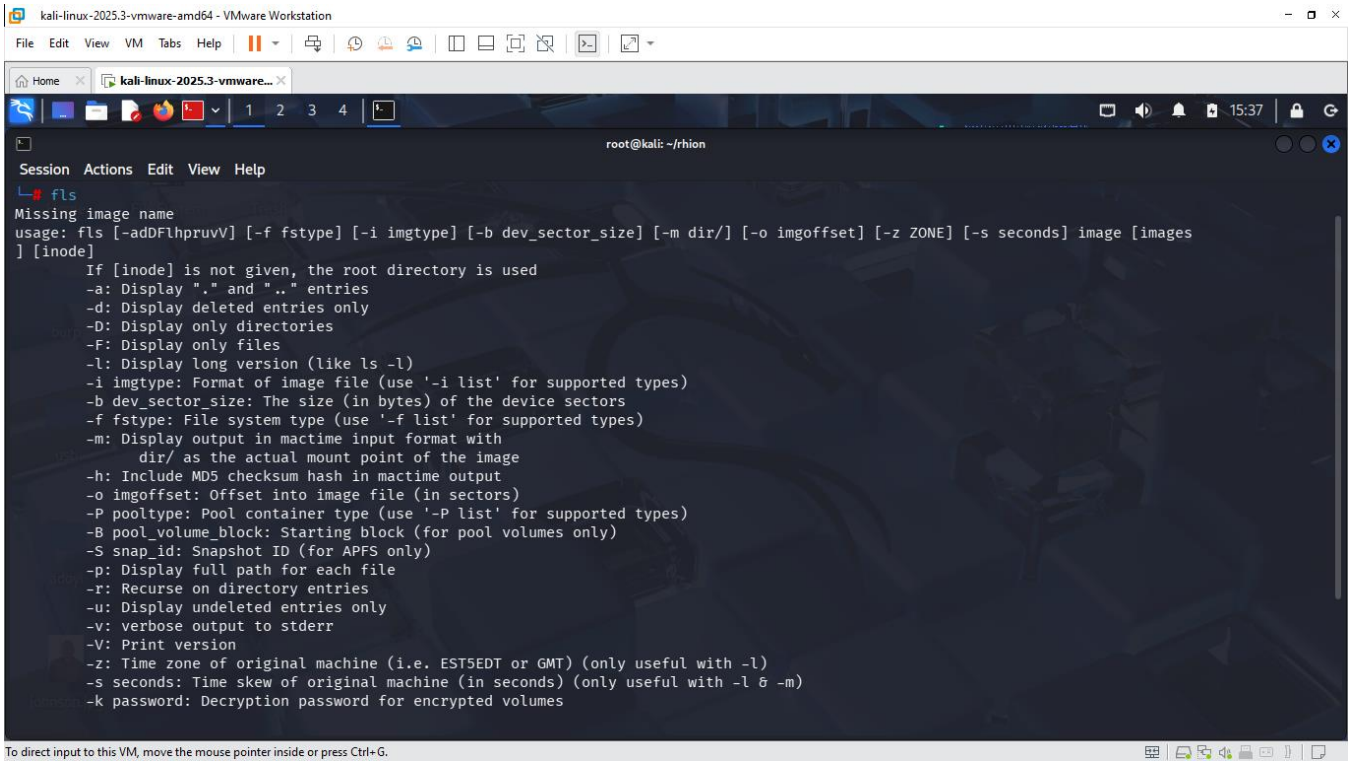

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.3-vmware...
14:46
root@kali: ~/rhion
Session Actions Edit View Help
┌──(root㉿kali)-[~/rhion]
└─# openssl dgst -md5 rhino.log
MD5(rhino.log)= c0d0093eb1664cd7b73f3a5225ae3f30
┌──(root㉿kali)-[~/rhion]
└─# openssl dgst -md5 rhino2.log
MD5(rhino2.log)= cd21eaf4acfb50f71ffff857d7968341
┌──(root㉿kali)-[~/rhion]
└─# openssl dgst -md5 rhino3.log
MD5(rhino3.log)= 7e29f9d67346df25faaf18efcd95fc30
┌──(root㉿kali)-[~/rhion]
└─# openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54
┌──(root㉿kali)-[~/rhion]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

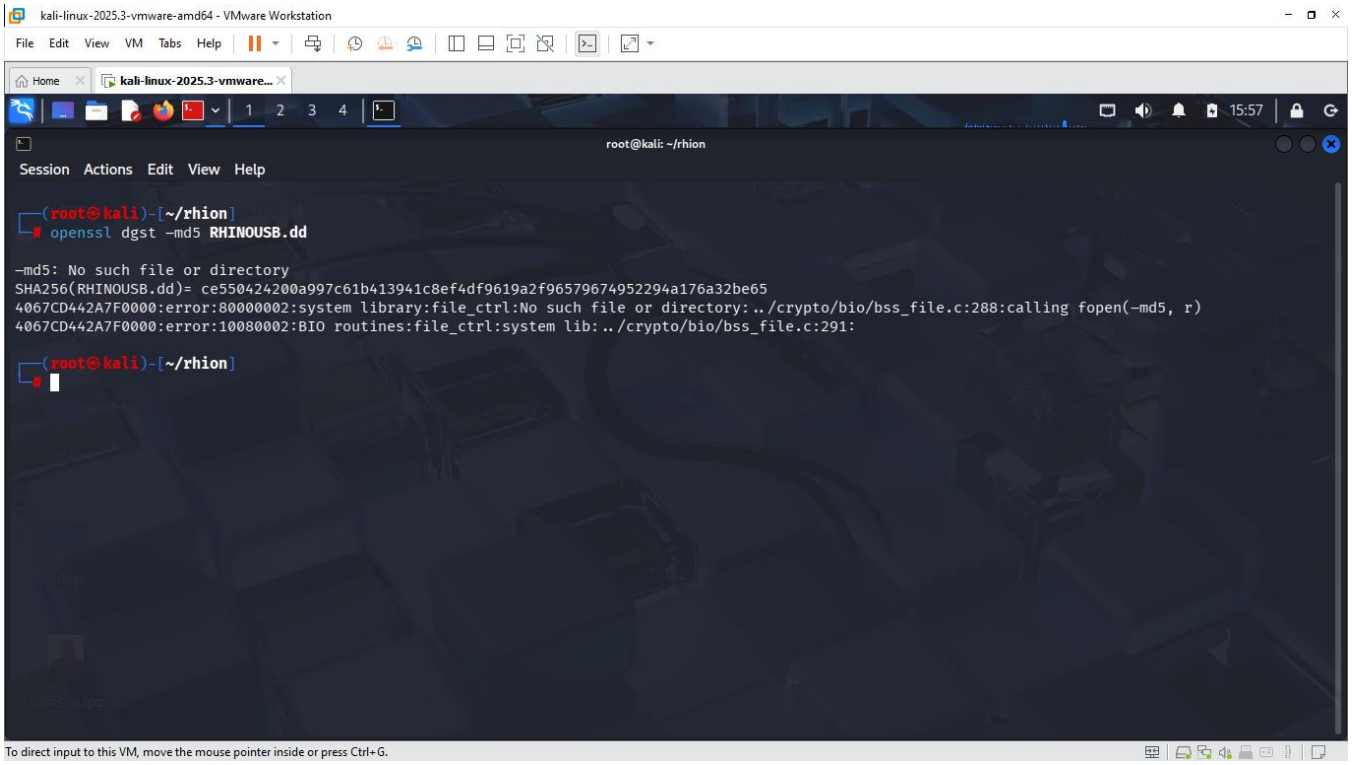

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.3-vmware...
15:00
root@kali: ~/rhion
CPU usage: 2.9%
Session Actions Edit View Help
┌──(root㉿kali)-[~/rhion]
└─# fdisk -l RHINOUSB.dd
Disk RHINOUSB.dd: 247.48 MiB, 259506176 bytes, 506848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x00000000
┌──(root㉿kali)-[~/rhion]
└─# mmls RHINOUSB.dd
┌──(root㉿kali)-[~/rhion]
└─# parted RHINOUSB.dd
GNU Parted 3.6
Using /root/rhion/RHINOUSB.dd
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) print
Model: (file)
Disk /root/rhion/RHINOUSB.dd: 260MB
Sector size (logical/physical): 512B/512B
Partition Table: loop
Disk Flags:
Number Start End Size File system Flags
1 0.00B 260MB 260MB fat16
(parted)
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

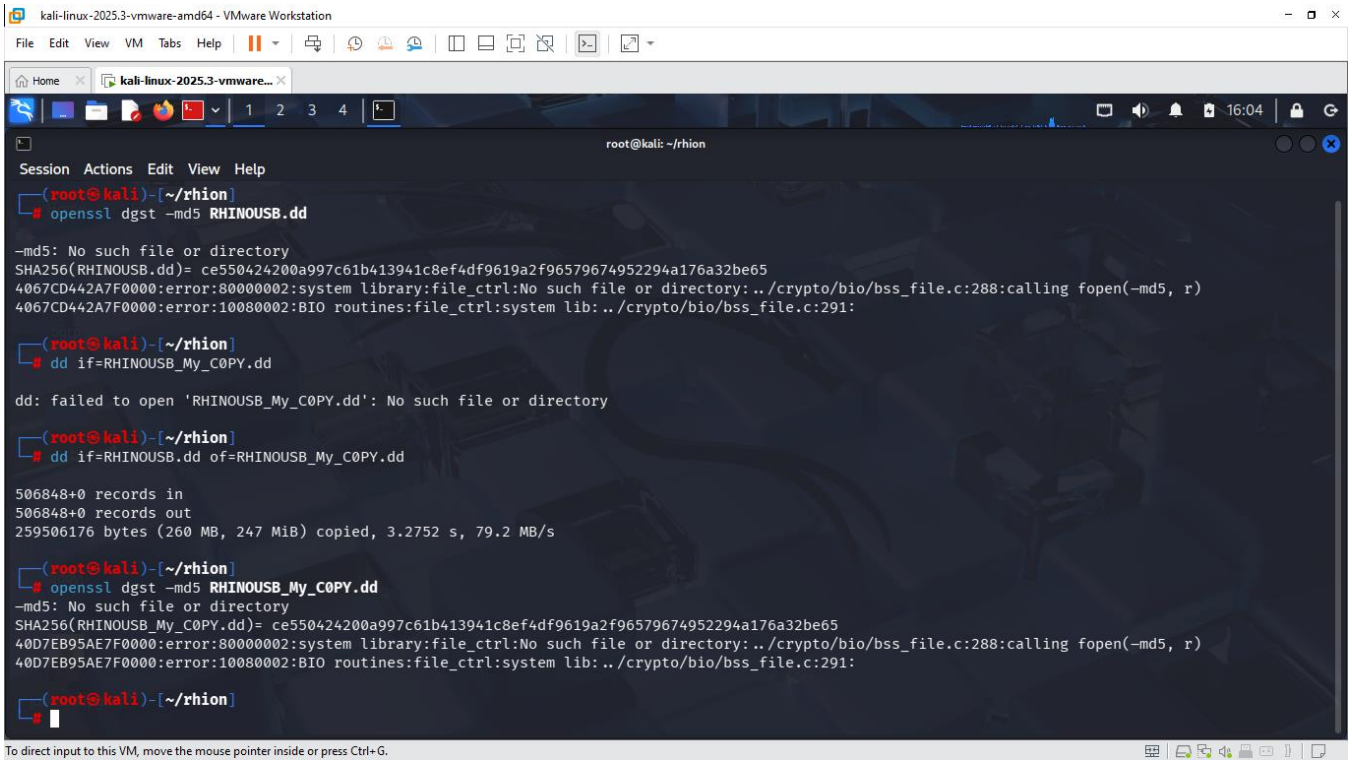

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Minimize
Home
kali-linux-2025.3-vmware...
15:29
root@kali: ~/rhion
Session Actions Edit View Help
┌──(root㉿kali)-[~/rhion]
└─# fsstate -b 512 -o 0 RHINOUSB.dd
Command 'fsstate' not found, did you mean:
command 'fsstat' from deb sleuthkit
Try: apt install <deb name>
┌──(root㉿kali)-[~/rhion]
└─# fsstat -b 512 -o 0 RHINOUSB.dd
FILE SYSTEM INFORMATION
File System Type: FAT16
OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847
METADATA INFORMATION
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


METADATA INFORMATION
Range: 2 - 8101622
Root Directory: 2
CONTENT INFORMATION
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 63290
FAT CONTENTS (in sectors)
529-536 (8) → EOF
537-544 (8) → EOF
┌──(root㉿kali)-[~/rhion]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Minimize
Home
kali-linux-2025.3-vmware...
15:34
root@kali: ~/rhion
Session Actions Edit View Help
┌──(root㉿kali)-[~/rhion]
└─# fls -o 0 RHINOUSB.dd
r/r 4: gumbo1.txt
r/r 6: gumbo2.txt
v/v 8101619: $MBR
v/v 8101620: $FAT1
v/v 8101621: $FAT2
V/V 8101622: $OrphanFiles
┌──(root㉿kali)-[~/rhion]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.3-vmware...
15:37
root@kali: ~/rhion
Session Actions Edit View Help
└─# fls
Missing image name
usage: fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/] [-o imgoffset] [-z ZONE] [-s seconds] image [images
] [inode]
If [inode] is not given, the root directory is used
-a: Display "." and ".." entries
-d: Display deleted entries only
-D: Display only directories
-F: Display only files
-l: Display long version (like ls -l)
-i imgtype: Format of image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-m: Display output in mactime input format with
dir/ as the actual mount point of the image
-h: Include MD5 checksum hash in mactime output
-o imgoffset: Offset into image file (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-S snap_id: Snapshot ID (for APFS only)
-p: Display full path for each file
-r: Recurse on directory entries
-u: Display undeleted entries only
-v: verbose output to stderr
-V: Print version
-z: Time zone of original machine (i.e. EST5EDT or GMT) (only useful with -l)
-s seconds: Time skew of original machine (in seconds) (only useful with -l & -m)
-k password: Decryption password for encrypted volumes
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.3-vmware...
15:57
root@kali: ~/rhion
Session Actions Edit View Help
(root㉿kali)-[~/rhion]
openssl dgst -md5 RHINOUSB.dd
-md5: No such file or directory
SHA256(RHINOUSB.dd)= ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65
4067CD442A7F0000:error:80000002:system library:file_ctrl:No such file or directory:../crypto/bio/bss_file.c:288:calling fopen(-md5, r)
4067CD442A7F0000:error:10080002:BIO routines:file_ctrl:system lib:../crypto/bio/bss_file.c:291:
(root㉿kali)-[~/rhion]
#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.3-vmware...
16:04
root@kali: ~/rhion
Session Actions Edit View Help
(root㉿kali)-[~/rhion]
openssl dgst -md5 RHINOUSB.dd
-md5: No such file or directory
SHA256(RHINOUSB.dd)= ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65
4067CD442A7F0000:error:80000002:system library:file_ctrl:No such file or directory:../crypto/bio/bss_file.c:288:calling fopen(-md5, r)
4067CD442A7F0000:error:10080002:BIO routines:file_ctrl:system lib:../crypto/bio/bss_file.c:291:
(root㉿kali)-[~/rhion]
dd if=RHINOUSB_My_C0PY.dd
dd: failed to open 'RHINOUSB_My_C0PY.dd': No such file or directory
(root㉿kali)-[~/rhion]
dd if=RHINOUSB.dd of=RHINOUSB_My_C0PY.dd
506848+0 records in
506848+0 records out
259506176 bytes (260 MB, 247 MiB) copied, 3.2752 s, 79.2 MB/s
(root㉿kali)-[~/rhion]
openssl dgst -md5 RHINOUSB_My_C0PY.dd
-md5: No such file or directory
SHA256(RHINOUSB_My_C0PY.dd)= ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65
40D7EB95AE7F0000:error:80000002:system library:file_ctrl:No such file or directory:../crypto/bio/bss_file.c:288:calling fopen(-md5, r)
40D7EB95AE7F0000:error:10080002:BIO routines:file_ctrl:system lib:../crypto/bio/bss_file.c:291:
(root㉿kali)-[~/rhion]
#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Investigation task

Conduct the examination in stages: preserve the supplied material; recover and validate relevant files from the disk image; examine media for hidden content; reconstruct and validate material FTP and HTTP objects; then correlate the artefacts in a single UTC timeline. Base every finding on identifiable records such as a source filename, hash, partition or metadata record, packet number, TCP stream, timestamp, response header, tool output, or screenshot. The questions below define the minimum scope of the report.

No	Investigation question	Minimum evidence to discuss
1	How did you preserve the case materials?	For each supplied disk image and capture, record source, original file name: **660*450-city.shot.jpg**, byte size, SHA-256 value, working-copy method, tool version, and file-access precautions.
2	What does the USB disk image show before recovery?	Document the partition layout, file-system information, allocated or unallocated area examined, relevant directory or metadata records, and any limits of file-system metadata.
3	What deleted or recoverable files are supported?	Describe the recovery method, output folder, recovered-file type, hash, visual or metadata check, and whether files are unique or duplicates. Do not report deck answers as your own findings.
4	Is there evidence of hidden content in the recovered media?	Record metadata review, steganography detection results, the authorised offline recovery method used, recovered output, hashes, and the evidence that supports or weakens a hiding interpretation.
5	What does the FTP capture establish?	Identify relevant sessions, transfer direction, packet or stream references, server response, object boundaries, reconstructed file validation, and timestamps. Treat any visible credential as sensitive evidence; do not reuse it.
6	What does the protected archive analysis establish?	Show that the reconstructed archive is complete, identify the evidence of encryption, and document any authorised course-only password-recovery workflow. Compare recovered content with other evidence using hashes where available.
7	What does the HTTP capture establish?	Identify material requests and responses, client and server endpoints, content type, stream or packet references, object extraction and validation. Do not execute any reconstructed executable or contact a recorded host.
8	What conclusion can you defend?	Build a UTC timeline, distinguish direct observations from inference, state a confidence level, and explain at least three limitations or alternative explanations concerning possession, transfer, and user attribution.

Technical starting points

The following are course-aligned starting points, not a substitute for documenting your own method. Work only on copies. Replace placeholders with values from your evidence and preserve the outputs that substantiate each decision.

Purpose	Course-aligned starting point
Preserve and identify evidence	Create read-only originals and working copies. Record SHA-256 values before and after copying; use file type and hex inspection only on working copies.
Examine the disk image	Use the Sleuth Kit workflow demonstrated in class to review media layout and file-system records (for example, mmls, fsstat, fls, and icat where appropriate).
Recover deleted files	Use the authorised PhotoRec or file-carving workflow on the working copy. Record the recovery scope selected and the output location; hash every material recovered file.
Review media and hidden data	Review file type and metadata, then use the supplied steganography workflow to detect and, if justified, recover embedded course content. Keep detector output and recovery notes.
Review FTP traffic	Use Wireshark display filtering and Follow TCP Stream to locate control and data sessions. Preserve packet or stream numbers, direction, timestamps, and raw object boundaries.
Review HTTP traffic	Use http, http.request, http.response, and tcp.stream eq <number> as starting filters. Confirm reassembly, response headers, content type, and file boundaries before exporting a working copy.
Validate reconstructed objects	Use magic number and trailer checks, file-type output, byte size, and SHA-256 hashes. A matching hash can establish binary equality but does not identify a person or original source by itself.

Evidence interpretation safeguards

- Use a hash and read-only preservation record for each original. A recovered working copy or carved file must remain traceable to its source and recovery method.
- For network objects, preserve the relevant packet range or TCP-stream reference, transfer direction, and reassembly method. A single packet is not necessarily a complete file.
- Visible FTP or email credentials are sensitive evidence. Record only what is necessary for assessment, redact them in screenshots where appropriate, and never reuse them outside the supplied course artefacts.
- An authorised password-recovery step may be used only against the supplied training archive and only within the course lab. Do not apply recovered passwords, wordlists, or methods to any other file, account, host, or service.
- A file hash, IP address, host name, username, or directory record can corroborate a relationship, but it does not alone prove original source, completion of an action, physical possession, or human identity.
- Do not execute a reconstructed executable, connect to a recorded endpoint, submit material to public scanners, or upload case evidence outside the approved lab. Follow lecturer-approved secure-disposal instructions only after assessment.

Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and case title.
2. Executive conclusion. State a concise, cautious conclusion and confidence level. Do not present a person-level attribution unless the artefacts support it.
3. Evidence acquisition and integrity. Record sources, filenames, sizes, hashes, read-only originals, working copies, tools, versions, and access controls.
4. Method. Explain the disk-image, recovery, steganography, FTP, HTTP, Wireshark, object-validation, and time-normalisation methods used.
5. Findings. Answer every investigation question with a precise evidence reference: file path, hash, metadata record, packet number, stream, timestamp, header, or screenshot ID.
6. Timeline and correlation. Present the material sequence in UTC. State how artefacts corroborate, conflict with, or fail to establish each proposed relationship.

7. Conclusion and limitations. Explain gaps, alternative explanations, duplicate files, partial captures, encryption, timestamp uncertainty, and the distinction between technical account evidence and a person at a keyboard.
8. Appendices. Include labelled screenshots, selected tool output, the completed timeline, the completed evidence log, and only the supporting material requested by your lecturer.

Timeline template

Complete a version of this table in your report or evidence appendix. Add rows for all material events and keep the main timeline in UTC.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Media preservation					
Disk image / file-system review					
Recovered or hidden content					
FTP transfer activity					
HTTP transfer activity					
Corroborated conclusion					

Evidence log template

Use an evidence log that lets an assessor locate the exact source file, tool, command or workflow, packet or stream, record, hash, and screenshot that supports each finding.

Evidence ID	Tool and method	Source record, stream, or path	Observed fact	Screenshot or appendix reference
E01				
E02				
E03				
E04				
E05				
E06				
E07				

Submission rules

- Work individually. Your analysis, report wording, screenshots, timeline, and evidence log must be your own work.
- Submit the report as SBTDF204_CaseStudy4_RegistrationNumber.pdf or .docx. If separate supporting material is requested, use SBTDF204_CaseStudy4_Evidence_RegistrationNumber.zip.

- Do not upload the original disk image, packet capture, recovered archive, media file, or executable unless your lecturer specifically requests it. Submit only the report and required supporting evidence.
- Use only the authorised, offline course artefacts in the approved lab environment. Do not contact people, accounts, hosts, or services; do not reuse credentials; do not execute recovered binaries; and do not publicly upload or share case evidence.
- Redact unrelated personal or sensitive data in screenshots while retaining enough packet, stream, path, timestamp, hash, and tool detail for the examiner to verify your finding.

Marking guide

The E-Campus maximum score applies to this case study. The assessor will apply the following percentage weighting to that maximum score.

Criterion	What strong work demonstrates	Weight
Evidence acquisition and recovery	Preserves the supplied material, documents integrity and working copies, and uses a defensible disk-image and deleted-file recovery workflow.	15%
Recovered-file and steganography analysis	Validates recovered media, records metadata and steganography results, and supports findings with clear tool output, hashes, and limitations.	20%
FTP and protected-archive analysis	Analyses FTP sessions, reconstructs and validates material objects, and documents the authorised course-only archive workflow without reusing credentials.	20%
HTTP and Wireshark traffic analysis	Uses request, response, stream, and packet evidence to extract and validate relevant objects while accounting for reassembly and transfer direction.	20%
Correlation, timeline, conclusion, and limitations	Builds a coherent UTC timeline, distinguishes observations from inferences, corroborates sources, and gives a cautious conclusion.	15%
Report quality and evidence presentation	Writes clearly, labels screenshots and output, cites evidence IDs consistently, and supplies a usable evidence log.	10%

Final check before submission

- Every major conclusion cites a source artefact, method, hash or packet/stream reference, timestamp, and screenshot or appendix reference.
- Your timeline has at least five material events across at least three artefact families and uses UTC consistently.
- Your report explains how deleted-file recovery, hidden-content analysis, FTP, and HTTP evidence corroborate or challenge the proposed sequence.
- Your conclusion distinguishes technical evidence of a file, transfer, account, or endpoint from proof of possession, transfer completion, and a named person.